Paul Eam

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EDUCATION

Northwestern University Evanston, IL
*B.S./M.S. in **Mechanical Engineering** (concentration in robotics),* Major GPA: 3.79/4 June 2027
Relevant coursework: Advanced Mechatronics, Advanced CAD, Machine Dynamics, Mechanics of Materials, Rendering, Hon. Eng. Analysis

WORK EXPERIENCE

Product Design Engineering Intern, *Upper Story LLC* June 2025- Sept 2025

- Led mechanical design of a consumer RC car, optimizing all components for injection molding; standardized fasteners and incorporated snap-fit features to streamline assembly and minimize labor costs
- Designed and implemented a custom SMT connector system for a snap-in battery system, enabling a 24% capacity increase and 49% increase in flywheel power
- Developed all CAD models and tooling strategies, reducing CNC machining time by 35.4% through optimized tool paths; fabricated proof-of-concepts using resin printers and CNC milling for rapid prototyping
- Engineered a battery charging station from the ground up with a focus on user interaction and manufacturability; incorporated LED charge indicators and USB-C input for modularity and ease of replacement

PROJECT EXPERIENCE

Lead Brakes Engineer, *Northwestern Formula Racing* Oct 2023- June 2024

- Designed front and rear brake rotors using SolidWorks, FEA, and Ansys Thermal Analysis; yielding a factor of safety of 2.24
- Changed material to 4130 steel and developed a unique rotor profile that enabled reliable RPM readings via Hall effect sensors while reducing debris buildup, becoming the standardized design for subsequent vehicles
- Calculated required specifications for calipers, brake bias, dimensions, etc. using MATLAB
- Manufactured the rotors using laser cutters, manual lathes, and CNC mills

Heat Sensor Mount Design, *Northwestern Formula Racing* Dec 2024- May 2025

- Collaboratively worked with the data acquisition team to design and implement Northwestern's first-ever tire temperature sensor mount; proven reliable through 15+ hours of driving
- Prototyped mounts using 3D printing and waterjet-cut steel, culminating in a cost-effective bent sheet metal design optimized for manufacturability
- Created GD&T documentation for three components for outsourced manufacturing

Design Thinking Communications I, II
Client: A Northwestern Faculty Member Jan 2024- Mar 2024

- Developed an attachable adapter for refrigerator water dispensers to make them more accessible for people with physical disabilities
- Designed and produced 5 functional prototypes iterations in a month
- Prototyped a 12°-actuated mechanism using varying plastics with distinct material properties to integrate both rigid and flexible components; designed in Inventor and 3D-printed in the Rapid Prototyping Lab

Client: Bellevue Performing Arts Center, *TN* Apr 2024- June 2024

- Led a team of 4 in creating a transportation solution for 150-lbs vinyl flooring with a device that weighs 20 lbs, specifically for small dance studios with limited staff
- Designed and built a mobile storage system using CNC routers, 3D printers, and hand-held tools
- 4 additional units requested by client after successful implementation
- Created a decrease in panel setup time by 45% based on user testing

Skills: Rapid Prototyping, DFM, DFA, Mechatronics, UX Design, Resourcefulness, Attention-to-Detail, User Research, Initiative
Certifications: Certified Solidworks Professional, Segal Design Certificate (2026)
Software Skills: CAD (Solidworks, Inventor, NX, Onshape), FEA, C, Python, MATLAB, Keyshot
Manufacturing: 3D-printing (FDM, resin, SLS), injection molding, soldering, GD&T, lathes, CNC routers/mills, waterjet, bandsaw

Project details can be viewed in my portfolio through this link: https://paulea1045.github.io/pe/index.html